

Supporting information: private memory and environmental recurrence

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This document accompanies the [research manuscript](#). It identifies what each comparison measures, which populations are independently prepared, and what a reader can reproduce from this compact release. All outcomes are from an abstract bit-string population model.

S1. What the findings establish

The original separate-introduction comparison supports opposite historical-minus-fresh carrier-frequency rankings under recurring and independent targets. Study 046 independently reproduces those two rankings and their direct interaction. Study 052 separately reproduces the later partial-recurrence direct-competition findings using independently prepared populations. These are distinct replication claims.

Studies 053-055 challenge other explanations using the study-052 genotype cohort. They change competitors, initial policy composition, and resident-policy preparation. New trajectories in those studies are not additional independent preparation cohorts. In particular, study 055 tests a policy substitution in a specific prepared population against the same unchanged founder. It does not generate memory storage from an initially memory-free individual: the fresh policy also carries an unused cache.

The 055 experiment and its saved-record reconstruction were complete before manuscript release preparation. The publication stage integrates the result and recomputes saved-table statistics. It introduces no new experiment, seed search, parameter search or outcome-dependent exclusion.

S2. Study and dependence map

Study	Scientific comparison	New work and reused evidence
045	One historical carrier versus one fresh-probe carrier, each against non-retrieving residents	768 new fresh paths; historical controls and targets from earlier studies
046	Independent test of the separate-introduction ranking reversal	2,304 new paths in an independent target/continuation cohort
047	Finite resident preparation versus matched naive starts	4,608 new transfers; all eligible prepared states from 046 retained
048	Historical probe versus a random probe at the same parent-cache distance	1,536 new paths; 4,608 authenticated controls from 047
049	Exact, partial and zero lag-two copying after recurrent preparation	2,304 new paths; 1,152 stored full-recurrence controls
050	Probabilistic versus deterministic global survival	3,456 new probabilistic paths; 3,456 deterministic controls
051	H and F in the same population versus separate introductions	4,608 new paths; 2,304 isolated controls; paired physical placements
052	Independent preparation replication of direct competition	192 new preparations and 6,912 new transfers; no old/new cohort pooling
053	Historical versus fresh focal policy under each fixed competitor, at constant carrier density	2,304 new HH/FF paths; 2,304 stored mixed controls from 052
054	Rare H among F, or rare F among H, with all individuals retrieving	2,304 new paths; genotype arrays and exogenous inputs reused from 052
055	One policy substitution versus the same unchanged founder after resident-policy preparation	1,152 new preparations and 3,456 new transfers; 192 source genotype arrays reused from 052

The identifiers are archival study identifiers, not a count of independent experiments supporting every claim. A reused control can appear in several tables without becoming new evidence. Each study's full numerical grid is in [STUDIES.json](#), which links its summaries, block aggregates and bootstrap rows.

S3. State, objective and candidate budget

There are 32 individuals and 32 binary coordinates. At each update the target is a 32-bit value. Raw loss is normalized Hamming mismatch; raw accuracy is one minus mean normalized mismatch. Strategy, founder identity and cache are distinct fields. A cache records one ancestral state and its provenance. In the two-policy experiments, H and F both maintain caches, but only H reads the stored state as its probe.

Each update evaluates 128 candidate slots: 32 current parents, 32 policy probes, 32 fresh scouts and 32 local descendants. Duplicate candidates are still charged. An H probe is its owner's cached genotype. An F probe is its current genotype XOR an independent uniform 32-bit mask. Scouts use the accepted fresh-variation rule for every parent. A local descendant is based on the best raw-scoring member of its parent's current/probe/scout set and flips each bit independently with probability 1/32. Zero and multiple flips are retained.

An unchanged surviving parent preserves its cache. A surviving nonparent carrier stores its recipient parent's previous current state, including when the candidate is a duplicate. Labels and founder identities follow the recipient parent. There is no spontaneous policy innovation or recombination.

The initial 32 evaluations plus 40×128 candidate evaluations yield 5,152 objective calls per stage. Study 055 has 4,608 stages, totaling 23,740,416 objective calls. Preparation and transfer calls are both counted. Equal objective-call budgets do not imply equal storage, runtime, diversity or biological energetic cost.

S4. Survival and cost

Earlier studies retain the 32 candidates with smallest penalized mismatch, using saved priorities and deterministic index fallbacks for ties. Probabilistic studies use weighted sampling without replacement. A candidate with penalized mismatch s has weight 2^{-s} , and its race key is $-\log(u) \times 2^s$. The 32 smallest keys survive, with saved priorities and indices breaking ties. Uniform values are constructed from stored 52-bit integers by $u = (k + 1)/(2^{52} + 1)$.

A carrier penalty of one mismatch bit is an explicit score convention, not a calibrated energetic cost. When only some individuals are carriers, it can alter their relative survival. When all candidates are carriers, the same one-bit penalty multiplies every race key by two and preserves the ordering; this is why studies 054-055 do not add a redundant common-cost arm. It does not overturn negative costly introductions against non-carriers.

Reported computational reconstruction permits at most two units in the last place for independently recomputed race keys while requiring exact selected indices and order. Study 055 observed differences no larger than one unit in the last place; genotype, score, ancestry and cache checks were exact. These are implementation checks, not a proof that a pseudorandom implementation samples an ideal continuous distribution.

S5. Environmental laws

The original recurrent law alternates two independently drawn uniform targets. The original independent law draws a fresh uniform target each update. Later experiments use ZERO, HALF and FULL to denote lag-two copy probabilities 0, 1/2 and 1. Under HALF, a target copies its own lag-two predecessor when its fixed copy bit says to copy; otherwise it uses a fresh innovation. A parity chain can therefore include renewed targets that are subsequently copied.

Unconditionally, adjacent targets are independent uniform under these laws. That identity does not imply that each realized adjacent distance is matched across the finite sample, or that the target is independent of the prepared population. Study 049 retains a previously recurrent preparation and first target pair. Study 055 instead applies the assigned law during both additional resident-policy preparation and continuation. Those different treatment contexts must not be pooled into a single recurrence response curve.

S6. Resident-policy intervention in study 055

Every one of the 192 study-052 genotype arrays is used. For each array, all-H and all-F populations begin with caches initialized to their own current states and undergo 40 additional updates under each law. No endpoint is excluded based on fitness, diversity or ancestry. The resulting 1,152 complete resident populations are frozen before transfer.

Genotypes and caches at transfer are preserved exactly. Founder labels are rebased to current physical positions, retaining a map to their preparation ancestry. Two fixed physical positions define paired single-policy substitutions. In each prepared background, the three paths are STAY, SWITCH0 and SWITCH1. One STAY path supplies both unchanged-founder controls. H-to-F changes only use of the cache; F-to-H activates an already present cache.

The target law continues across the stage boundary using the final two preparation targets. Transfer uses 1,200 prospectively fixed continuation seed records across innovation, copy and continuation families. New seeds and new paths do not remove dependence on the reused source genotype cohort.

For a given background, SWITCH is the mean of the introduced founder's outcomes in the two switched placements. STAY is the mean of those same two founders' outcomes in the unchanged population. $EFFECT = SWITCH - STAY$ is formed within continuation and then averaged within block. Separate absolute SWITCH changes answer whether the introduced type increased in mean, rather than whether it exceeded its matched control.

All five fixed partial-recurrence criteria are supported individually: positive F-to-H EFFECT, negative H-to-F EFFECT, positive absolute H introduction, negative absolute F introduction, and positive HALF-minus-ZERO F-to-H EFFECT. These pointwise results do not confer simultaneous confidence on their conjunction. ZERO leaves F-to-H unresolved and supports H-to-F benefit. Under HALF, both terminal population-accuracy effects remain unresolved.

S7. Endpoints and uncertainty

Quantity	Meaning
F0, F1, F40	Founder or policy frequency initially, after one update and after 40 updates, as identified by the series key
D40	$F40 - F0$; this differs from final frequency when initial frequencies differ
F	Mean post-selection frequency across the 40 updates
FE1-FE8	Mean frequencies within the eight fixed five-update windows
U0, U1, U40	Initial, first-update and terminal raw population accuracy
U	Mean post-selection raw population accuracy across 40 updates
E1-E8	Accuracy over the eight fixed five-update windows
MLOSS	Mean post-selection normalized mismatch across the 40 updates; equal to $1 - U$

Complete keys also identify study-specific regime, cost, policy, background and comparison. Read a key together with that study's analysis.py and config.json; similarly named series across studies need not have identical intervention contexts.

Eight paired continuations are averaged inside each of 24 independent target blocks. The stored 2,000 bootstrap rows resample whole blocks. Means and 2.5th/97.5th percentiles of resampled means use NumPy's linear quantile method. The intervals are approximate and pointwise across a large correlated grid. No family-wide error guarantee is claimed. A positive interval supports a positive mean for that comparison; an interval containing zero remains unresolved and does not establish equivalence.

Structural zeros follow an intervention identity. Observed exact zeros describe this finite sample. Their degenerate intervals do not prove a zero population probability. Likewise, the reported zero-event upper bound concerns any survivor among eight continuations in a new block, not a per-lineage long-run fixation probability.

S8. Contrary outcomes and local mathematics

The full grids retain all costs, laws, windows and predefined comparisons. They preserve negative changes, endpoint disagreements, ties and missing crossings. In study 055, the partial-recurrence H-to-F second placement contains a fresh-founder fixation at block 8, continuation 1, update 14; the negative mean does not erase that event. Study 054's sampled extinction of every rare F introduction is not a theorem of impossible establishment.

The [local candidate calculation](#) and [fixed-pool counterexample](#) distinguish pairwise distance from whole-pool selection. The [parity-chain note](#) describes an ideal target law. The [cost sensitivity](#) and [quantization notes](#) have explicit mathematical assumptions. They are not population observations and do not prove biological generality or certify floating-point logarithms and pseudorandom generators.

S9. Public reproduction and raw-data boundary

From the repository root:

```
python -m pip install -r requirements.txt
python scripts/verify_release.py
python scripts/reproduce_statistics.py
python scripts/rebuild_figures.py
```

The statistics check recomputes all 14,729 intervals from the included block aggregates and exact saved bootstrap indices. It also verifies every registered manuscript/figure statistic against its source summary. Figure generation uses those same summaries. Both write to `_rebuilt/` by default and neither runs a population experiment.

Archived scientific modules in `code/045` through `code/055` preserve each study's operators and analysis. Site-specific Slurm and SSH launch scripts are omitted. Some archived modules require original input tapes, directory layouts or predecessor packages; they are provided for inspection, not advertised as a turnkey rerun interface. Do not infer successful raw-trajectory reproduction from a passing saved-table check.

The multi-gigabyte candidate-by-candidate paths, random tapes and complete execution logs remain in the project archive and are not included in this compact public package. The archived independent reconstruction receipts describe checks already performed against those files. The public source map retains original and exported hashes and labels administrative-path transformations. Public integrity manifests authenticate this release's files, not files that are absent from the release.

S10. Research status

This preprint synthesizes a bounded computational result. Internal agent review is distinct from external peer review, and correct computation is distinct from independent validation of a model. Practical effect size, applicability beyond this model, the origin and maintenance of memory storage, equilibrium invasion and general costly advantage remain open. The experiment schedules remained paused while the manuscript was prepared.